
```
def list_random(n1,n2,n3):  
    output = [1]  
    while len(output) <=n3:  
        output.append(next_random(output[len(output) - 1],n1,n2,n3))  
    return(output)
```

Consider the list we get by running `list_random(29,23,32)`:

```
[1, 20, 27, 6, 5, 8, 31, 26, 9, 28, 3, 14, 13, 16, 7, 2, 17, 4, 11, 22, 21, 24,  
15, 10, 25, 12, 19, 30, 29, 0, 23, 18, 1]
```

It's not easy to detect a simple pattern in this list, which is exactly what we wanted. One thing we can notice is that it contains only numbers between 0 and 32. We may also notice that this list's last element is 1, the same as its first element. If we wanted more random numbers, we could extend this list by calling the `next_random()` function on its last element, 1. However, remember that the `next_random()` function is deterministic. If we extend our list, all we would get is repetition of the beginning of the list, since the next "random" number after 1 will always be 20, the next random number after 20 will always be 27, and so on. If we continued, we would eventually get to the number 1 again and repeat the whole list forever. The number of unique values that we obtain before they repeat is called the *period* of our PRNG. In this case, the period of our LCG is 32.

Judging a PRNG

The fact that this random number generation method will eventually start to repeat is a potential weakness because it allows people to predict what's coming next, which is exactly what we don't want to happen in situations where we're seeking randomness. Suppose that we used our LCG to govern an online roulette application for a roulette wheel with 32 slots. A savvy gambler who observed the roulette wheel long enough might notice that the winning numbers were following a regular pattern that repeated every 32 spins, and they may win all our money by placing bets on the number they now know with certainty will win in each round.

The idea of a savvy gambler trying to win at roulette is useful for evaluating any PRNG. If we are governing a roulette wheel with true randomness, no gambler will ever be able to win reliably. But any slight weakness, or deviation from true randomness, in the PRNG governing our